MILTARY SPECIFICATION

LASER, GUNNERY TRAINER: M55

WARNING EYE HAZARD

THE LASER BEAM CAN BE DANGEROUS AND CAUSE BLINDNESS IF IT ENteRS THE EYES EITHER DIRECTLY OR REFLECTED OFF OF A REFLECTED SURFACE. BEFORE FIRING THE LASER, OPERATOR SHOULD TAKE ADEQUATE PRECAUTIONS TO ASSURE MAXIMUM SAFETY

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of laser device, Designated Laser, Gunnery Trainer: M55, with equipment, Part number 10554640. The Laser Gunnery Trainer: M55 without equipment iS Part Number 11741700.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-O-13830	Optical Components For Fire Control Instruments, General Specification Governing The Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and inspection of
MIL-I-45607	inspection Equipment; Acquisition Maintenance, and Disposition of

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-sTD-810	Environmental Test Methods
MIL-STD-1275	Electrical Circuit, 28 Volt DC Transient Characteristics For Military Vehicles.

DRAWINGS

U.S. Army, Frankford Arsenal

F8565556	Pressure Tester
A10554640	Laser, Gunnery Trainer: M55 W/E
F11739405	Final Test Fixture For Laser Trainer: M55
F11739406	Fixture, Collimation and Focus
F11741700	Laser, Gunnery Trainer: M55
F11743778	Case, Carrying Assembly
A11746797	Tube, Laser, Plasma, HE-NE

PACKAGING

PS11741700	Packaging of Laser, Gunnery Trainer, M55
PS1055464O	Packaging of Laser, Gunnery Trainer: M55 W/E

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Fabrication. - The unit shall be manufactured in accordance with Drawing F11741700 and all drawings pertaining thereto.

3.2 General specification. - The laser trainer shall meet the following requirements, where applicable, of MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T19:25Z
Check the source to verify that this is the currentversion before use.

d. Electrical and electronic assemblies
e. Part identification and marking
f. Workmanship
g. Source control drawings
h. Applicable Quality Assurance Provisions

3.2.1 Assemblies. - Assemblies listed below shall have been inspected and accepted in accordance with their respective drawings and specifications prior to being assembled into the laser trainer.

a. Power Supply Module, Drawing F11739861
b. Timing Module, Drawing F11737703
c. Plasma Tube, Drawing Al1746797

3.3 Performance.

3.3.1 Interchangeability.- The laser trainer shall meet all the requirements of this specification without need for selective assemblies for any item. Unless otherwise specified in detail drawings or assemblies, all components shall be interchangeable. No incompatibility shall exist between the replacement Item and the mechanical interface/fits, or electrical and optical functions of the system.

3.3.1.1 Mechanical and electrical interface.- The dimensions of the laser trainer mounting ring shall be as specified on Drawing F11741676. The conductivity between the mounting ring and the mating portion of an M73 Machine Gun Mount, or suitable simulation, shall be such that DC resistance shall not exceed .01 ohm.

3.3.2 Reliability. - The reliability of the laser trainer shall be 200 hours mean time between failure (MTBF). This MTBF consists of 80 hours in the continuous mode of operation and 120 hours in the flash mode of operation. (The flash mode is defined as 1 flash per 3 seconds for 2 minutes followed by a 30 second rest).

3.3.3 Electromagnetic interference.- E.M.I. emission and susceptability of the laser trainer shall meet the requirements of CE03, CS02, CS06 and RE02 in accordance with MIL-STD-461 and voltage transients limits specified in MIL-STD-1275 for the normal vehicle system including batteries and generators.

3.3.4 Environmental Requirements.

3.3.4.1 Operating conditioner. - The laser trainer shall show no evidence of damage or deterioration and shall meet all requirements of 3.3.5 under the following environmental conditions.

3.3.4.1.1 Temperature. - After thermal stabilization at temperatures of O°F and 100°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

3.3.4.1.2 Humidity. - Conditions specified in Method 507.1, Procedure II, MIL-STD-81O, except the number of humidity cycles will be limited to one.

3.3.4.1.3 Altitude. - Conditions specified in Method 500.1, procedure I, MIL-STD-81O, exe the chamber pressure will be limited to 10,000 feet.

3.3.4.2 Non-operating conditions. - The laser trainer in the non-operating, de-energized condition, shall show no evidence of damage or deterioration and shall meet all requirements of 3.3.5 after the following environmental exposure.

3.3.4.2.1 Storage temperature. - Thermal stabilization at -65°F for 24 hours and +155°F for 24 hours.

3.3.4.2.2 Vibration "A". - Conditions specified in Method 514.2, Procedure VIII, Curve V, Figure 514.2-6, MIL-STD-810 (except that the resonance dwell time is to be deleted), with the laser trainer mounte horizontallY in an M73 Machine Gun Mount or equivalent simulation.

3.3.4.2.3 Vibration "B". - Conditions specified in Method 514.2, Procedure X, Curie AX, Figure 514.2-7, MIL-STD-810, with the laser trainer properly stored in Case, Drawing No. F11743778.

3.3.4.2.4 Shock. - Conditions specified in Method 516.2, Procedure I, figure 516.2-1, shock parameters (a) and (c) and Procedure II (limit to 30 inches and V, MIL-STD-810.

3.3.4.2.5 Altitude.- Conditioner specified in Method 500.1, Procedure I, MIL-STD-81O.

3.3.4.2.6 Sunshine.- Conditions specified in Method 505.1, Procedure I, MIL-STD-81O.

3.3.4.2.7 Rain.- Conditions specified i nMethod 506.1, Procedure I, MIL-STD-81O.

3.3.4.2.8 Humidity.- Conditions specified in Method 507.1, Procedure IV, Figure 507.1-3, MIL-STD-810,

3.3.4.2.9 Salt fog- Conditions specified in Method 509.1, Procedure I, MIL-STD-81O.

3.3.4.2.10 Dust.- Conditions specified in Method 510.1, Procedure I, MIL-STD-81O.

3.3.4.2.11 Fungus.- Conditions specified in Method 508.1, Procedure I, MIL-STD-81O.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T19:25Z
Check the source to verify that this is the current version before use.

3.3.5 Performance. - Operating current for the laser trainer shall be a maximum of 3 amps with 24 VDC applied to determine compliance with the following requirements.

3.3.5.1 Power input control. - The laser light shall be emitted in either a flash uncle or a continous mode.(see 3.2.5.2) when an external "depress to close" type of trigger switch, connected in series with the positive power connector, the mounting ring and a power source of 24 volts ± 6 volts, is actuated.

3.3.5.2 Continuous or flash mode switch.- The position of the mode switch shall properly, indicate whether the laser trainer is in a "continuous" or "flash" mode, as specified in 3.3.5.5 and 3.3.5.6.

3.3.5.3 Safe and fire laser switch.- In the "Safe" position, the "safe and Fire" switch shall positively inhibit the generation of laser light.

3.3.5.4 Trgger Lamp.- This lamp shall light when the external trigger switch spscified in 3.3.5.1 is depressed.

3.3.5.4.1 Laser "ON" Lamp.- This lamp shall light when the trainer is lasing in either the continuous or flash mode.

3.3.5.5 Continuous mode. - When the mode select switch is in the continuous mode position, depression of the external trigger switch shall cause the trainer to emit laser light as the the trigger is depressed. laser light shall be emitted within 100 msec after depression of the trigger.

3.3.5.6 Flash mode. - When the mode switch is in the "flash" position and external power is provided as specified in 3.3.5.1, depression of the external trigger switch shall cause the trainer to emit laser light for a period of 100 ± 25 millisec within 100 millisec of the depression of the trigger switch unless the trigger switch is released. There shall be no further laser light emission unless the external trigger switch is released and subsequently depressed.

3.3.5.7 Repetition rate.- Repeat operation of the flash or continuous mode shall be possible within 200 msec after the external trigger switch is released.

3.3.5.8 Power output.

3.3.5.8.1 The laser light power output shall be no less than 0.5 mini-watts and no greater than 2.0 milli-watts when measured at an ambient temperature of 72° to 78° and when measured at 100°F.

3.3.5.8.2 The laser light power output shall be within 0.2 to 2.0 mini-watts when measured at 0°F.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

3.3.5.9 Operating input voltage. - The laser trainer shall operate and meet the requirements of this specification when a voltage of 24 ± 6 VDC is applied as specified in 3.3.5.1 and shall cause laser light to be emitted as follows:

3.3.5.9.1 Thirty seconds or less between shots.- When the trainer has been fired within the previous 30 seconds, light emission shall occur within 100 milliseconds.

3.3.5.9.2 Thirty seconds to 3 minutes between shots.- When the Trainer has been fired within the previous 30 seconds to 3 minutes, light emission shall occur with 750 milliseconds.

3.3.5.9.3 Over 3 minutes between shots.- When the trainer has not been fired within the previous three minutes or longer time internal, light emission shall occur within 30 seconds.

3.3.6 Sealing. - The laser shall be capable of retaining an internal pressure of dry nitrogen at 5 ± 0.1 psig for a minimum of two hours without loss in pressure in excess of 0.2 pound.

3.3.6.1 Internal pressure. - Subsequent to meeting the sealing requirements of 3.3.6 all units shall be dry flushed with an internal atmosphere containing nitrogen having a dew point of at least -25°F. The internal pressure of the laser trainer shall be established at 1 psi ± 0.2 psi.

3.3.7 Collimation. - The laser beam shall be aligned parallel to the longitudinal axis of the mounting ring within 6 mils when measured at a target of 200 ± 20 feet.

3.3.8 Reverse polarity protection. - The laser trainer shall substain voltage input polarity reversal without damage.

3.3.9 Multi-purpose carrying case. - The multi-purpose carrying case shall meet all the requirements listed on Drawing F11743778.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of four trainer assemblies plus four each of all items covered by SQAP. One sample trainer shall be subjected to the fungus test (4.6.4.2.11), humidity non-operating (4.6.4.2.8) and salt fog (4.6.4.2.9). The remaining 3 samples shall be subjected to all other inspections and tests herein. The sample shall be manufactured In the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Cotnractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor’s inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be accordance with “Submission of Product” provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provision (SQAP) listed in the technical data package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.1.1 Certification.- Prior to assembly, the laser plasma tube, high voltage power supply, and the timing module shall be certified to the requirements and tests in their respective specifications. Certifi-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

cation shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

4.3.2.2 Trainer assembly.

4.3.2.2.1 First article.- First article testing shall be conducted in accordance with 4.2, 4.3.2.1 and Tables I through V.

4.3.2.2.2 100% inspection.- The requirements and related tests in Table I shall be subjected to 100% inspection.

TABLE I - 100% INSPECTION

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101. Mechanical and Electrical Interface	3.3.1.1	4.6.1.3
102. Continuous and Flash Mode	3.3.5.1	4.6.5.1
103. Continuous and Flash Mode Switch	3.3.5.2	4.6.5.2
104. Safe/Fire laser switch	3.3.5.3	4.6.5.3
105. Trigger lamp and laser "ON" LAmp	3.3.5.4 & 3.3.5.4.1	4.6.5.4
106 Continuous Mode	3.3.5.5	4.6.5.5
107. Flash Mode	3.3.5.6	4.6.5.6
108. Repetition Rate	3.3.5.7	4.6.5.7
109. Power Output	3.3.5.8.1	4.6.5.8
110. Input Voltage	3.3.5.9, 3.3.5.9.1, 3.3.5.9.2, 3.3.5.9.3	4.6.5.9
111. Sealing and Internal Pressure	3.3.6 & 3.3.6.1	4.6.6
112. Collimation	3.3.7	4.6.7
113. Reverse Polarity Protection	3.3.8	4.6.8
114. Multi-purpose Carrying Case	3.3.9	4.6.9
115. Interchangeability gauges	3.3.1	4.6.1.1

NOTE: These tests shall be conducted at a temperature between 60° and 90°F.

4.3.2.2.3 Environmetnatl. - Three trainer assemblies shall be selected at random as samples from each 25 produced, or from each month's production whichever occurs first. Each sample shall meet the requirements and tssts in Table II and shall then meet all the requirements and tests in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401.	Operating temperature	3.3.4.1.1	4.6.4.1.1
402.	Storage temperature	3.3.4:2.1	4.6.4.2.1
403.	Vibration "A" (w/o carrying case)	3.3.4.2.2	4.6.4.2.2
404.	Vibration "B"(w/carrying case)	3.3.4.2.3	4.6.4.2.3

4.3.3 Sontrol tests. - Three trainer assemblies shall be selected and meet the requirements and tests of Table III. These tests shall be conducted as follow-up tests to the first article samples at 25%, 50% and 75% points of completion of the production contract. One trainer shall be subjected to fungus (4.6.4.2.11) humidity non-operating (4.6.4.2.8) and salt fog (4.6.4.2.9). The remaining two trainers shall be subjected to all other tests In Table III.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
405.	Interchangeability	3.3.1	4.6.1.2
406.	Electromagnetic interference	3.3.3	4.6.3
407 l	Humidity - operating	3.3.401.2	4.6.4.1.2
408.	Altitude - operating	3*3*4.1.3	4.6.4.1.3
409.	Shock	3.3.4.2.4	4.6.4.2.4
410.	Altitude - non-operating	3.3.4.2.5	4.6.4.2.5
411.	Sunshine	3.3.4.2.6	4.6.4.2.6
412.	Rain	3.3.4.2.7	4.6.4.2.7
413.	Humidity - non-operating	3.3.4.2.8	4.6.4.2.8
414.	Salt fog	3.3.4.2.9	4.6.4.2.9
415*	Dust	3.3.4.2.10	4.6.4.2.10
416.	Fungus	3.3.4.2.11	4.6.4.2.11

4.3.4 General. - One assembly shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table IV.

TABLE IV

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
417.	Fabrication	3.1	Applicable Drawing-Visual
418.	General specifications	3.2	MIL-F-13926, MIL-O-13830 - visual

NOTE: These tests shall be conducted at a temperature between 60° and 90°F.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T19:25Z
Check the source to verify that this is the current version before use.

4.4 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the Contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP'S forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the Technical Data Package List.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment Is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment. shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Interchangeability. - The trainer shall be tested by means of interface gages and by interchange of assemblies as specified below to determine compliance with 3.3.1.

4.6.1.1 Interchangeability gaging.- The assemblies listed below shall be inspected by utilizing the gages specified respectively:

Assembly	Gage Drawing
a. Barrel Assembly, F11741701	B11739396 B11739397 311739398 B11739403

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:25Z
Check the source to verify that this is the current version before use.

Assembly	Gage Drawing
b. Electronic Assembly, Fl1737724	C11739399 C11739400

4.6.1.2 Interchange of assemblies/components.- Four laser trainers shall be selected to perform the interchangeability tests. The four trainers shall be dis-assembled for interchange of the following assemblies/components:

a. Lens Guard, C11741679
b. Cell Assembly, D11741680
c. Timing Module, F11737703
d. Power Supply, F11739861
e. Barrel Extension, C11741689
f. Laser Tube Assembly, Al1746797

4.6.1.2.1 Selection.- The four dis-assembled trainers shall be numbered (1 to 4) and presented to the Government Inspector for selection and re-assembly of four laser trainers as follows:

a. Trainer No. 1.- This trainer shall be assembled by randomly selecting one of each of the items listed in 4.6.1.2 from the four dis-assembled trainers.

b. Trainer No's 2,3 and 4.- These trainers shall be likewise assembled by randomly selecting from the remaining assemblies/components of 4.6.1.2.

4.6.1.2.2 Interchangeability test.- Subsequent to assembly of the four trainers resulting from 4.6.1.2.1 each trainer shall be subjected to the tests in 4.6.5, 4.6.6 and 4.6.7 to determine compliance with their respective requirements.

4.6.1.3 Mechanical and electrical interface.- This test shall be performed using Final Test Fixture, F11739405. A standard resistance bridge or equivalent shall be utilized to measure the DC resistance as specified in 3.3.1.1.

4.6.2 Reliability. - A reliability demonstration test shall be conducted as part of First Article Test (FAT) to assure compliance with the requirement of paragraph 3.3.2 at a 90% confidence level. A minimum of 4 hours of continuous mode operation and 6632 flashes in the flash mode on each of the four first article samples (see 4.2) shall be accumulated after/during exposure to each non-operating/operating environments of paragraphs 3.3.4.1 and 3.3.4.2. Additional operaton in the continuous and flash modes may be required at 75° ± 15°F in order to accumulate a minimum of 50 hours in the continuous mode and 66,312 flashes in the flash mode on each of the four first article samples. The trainers shall be tested until the total hours in the continuous mode and the total flashes in the flash mode meet the accept/reject criterial in Table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

NOTE: During continuous mode operation, the trainer shall be "ON" for 50 minutes and "OFF" for ten minutes each hour. The total "ON" time shall be accumulated to satisfy the operating hours as identified above.

TABLE V

CONTINUOUS MODE HOURS MINIMUM	FLASH MODE FLASHES MINIMUM	RELIABILITY FAILURES	
		ACCEPT NO.	REJECT NO.
185	265,248	0	7
312	837,020	1	7
426	1,145,402	2	7
535	1,437,751	3	7
640	1,720,201	4	7
742	1,995,872	5	7
843	2,266,486	6	7

4.6.2.1 Data collection. - Data collected will consist of all equipment malfunctions, number of hours and lasings completed when malfunction occurred, nature and cause of malfunction, time required to correct malfunction, required corrective action, item serial number and environmental exposure.

4.6.2.2 Failure definition. - A failure is defined as any malfunction which renders the laser trainer- inoperative or operative outside the limits of this specification. In determination of chargeable reliability failures the following clarification shall apply:

a. If upon failure analysis (to determine the cause of the malfunction) as assignable cause (related to manufacturing process, procedure, basic design weakness, etc.) is determined as the basis of the malfunction, the malfunction will not be chargeable as a reliability failure. However the correction to the manufacturing process or procedure (et al) shall be accomplished.

b. If upon failure analysis no assignable cause is evident and the malfunction is considered random in nature, the malfunction shall be a "changeable" reliability failure and as such must be considered in the above demonstration test criteria. The decisions as to chargeable reliability failure must be coordinated with the procuring activity, Frankford Arsenal Reliability and Maintainability Engineering Branch SARFA, Q4100.

4.6.2.3 System interruption.- A system interruption is any system malfunction for which the laser trainer is operable within the limits set forth in their specification, but requires maintenance to correct the malfunction. Although a system interruption does not penalize the reliability, it may constitute a failure of some major subassembly or component for which system acceptance cannot be made until that characteristic or component which caused the system interruption has been screened. Subsequent

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

to the reliability test, the sample systems shall be refurbished and readied for delivery and shipment.

4.6.3 Electromagnetic interference.- Electromagnetic measurements shall be performed in accordance with the applicable procedures specified in MIL-STD-462 for Class IC equipment to determine compliance with 3.3.3.

4.6.4 Environmental requirements.

4.6.4.1 Operating conditions. - With laser trainer operating in accordance with 3.5.5, there shall be no evidence of damage or deterioration and the trainer shall meet all the tests of 4.6.5 during the following environmental conditions to determine compliance with 3.3.4.1.

4.6.4.1.1 Temperature. - The test apparatus shall be in accordance with the applicable Test Facilities Provisions of MIL-F-13926, except that assemblies may be mounted on the inside surface of the test chamber. Perform this test in accordance with temperature cycling, Procedure I of MIL-F-13926, except with conditions of O°F and 100°F to determine compliance with 3.3.4.1.1 and the sequence of testing is optional. Temperature change shall not be greater than 3°F per minute. Allow temperature to stabilize for at least two hours. Then allow five minutes warm-up time and fire three times before performing test of 4.6.5. This test may be performed in conjunction with the test in 4.6.4.2.1.

4.6.4.1.2 Humidity. - This test shall be performed as specified in MIL-STD-81O, Method 507.1, Procedure II, while operating laser trainer to determine compliance with 3.3.4.1.2.

4.6.4.1.3 Altitude. - This test shall be performed as specified in MIL-STD-81O, Method 500.1, Procedure I, except that the chamber pressure will be limited to 10,000 feet to determine compliance with 3.3.4.1.3.

4.6.4.2 Non-operating conditions.- The laser trainer in the non-operating, de-energized condition, shall show no evidence fo damage or deterioration and shall be visually inspected for physical damage and then tested to meet 4.6.5 after the following environmental exposure to determine compliance with 3.3.4.2.

4.6.4.2.1 Storage temperature.- The test apparatus shall be in accordance with the applicable test facilities provisions of MIL-F-13926 except that assemblies may be mounted on the inside surface of the test chamber. To determine compliance with 3.3.4.2.1, conduct this test in accordance with Temperature cycling, Procedure I of MIL-F-13926, with conditions of -65°F and +155°F and the sequence of testing is optional. Temperature change shall not be greater than 3°F per minute.

4.6.4.2.2 Vibration “A”. - To determine compliance with 3.3.4.2.2, perform this test in accordance with Method 514.2 of MIL-STD-81O, procedure VIII, Curve V, Figure 514.2-6, with a vehicle mileage of 3000 miles. Delete resonance dwell time. The laser trainer shall be mounted horizontally in the Final Test Fixture, F11739405.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T19:25Z
Check the source to verify that this is the current version before use.

4.6.4.2.3 Vibration "B". - To determine compliance with 3.3.4.2.3, perform this test in accordance with Method 514.2 of M1L-STD-810, Procedure X, Curve AX, figure 514.2-7 for four cycles. Delete resonance dwell time. The laser trainer shall be stored in its Carrying Case, F11743778 for this test.

4.6.4.2.4 Shock. - To determine compliance with 3.3.4.2.4, this test shall be performed as specified MIL-STD-810, Procedure I, Figure 516.2-1, Shock parameters (a) and (c), Procedure II (limited to a 30 inch drop) and Procedure V. The laser trainer will be stored in its carrying case for Procedure II.

4.6.4.2.5 Altitude. - This test shall be performed as specified in MIL-STD-810, Method 500.1. Procedure I to determine compliance with 3.3.4.2.5.

4.6.4.2.6 Sunshine. - This test shall be performed as specified in MIL-STD-810, Method 505.1, Procedure I to determine compliance with 3.3.4.2.6.

4.6.4.2.7 Rain. - This test shall be performed as specified in MIL-STD-810, Method 506.1, Procedure I to determine compliance with 3.3.4.2.7. Also there shall be no water present in the interior of the trainer.

4.6.4.2.8 Humidity. - This test shall be performed as specified in MIL-S-810, Method 507.1, Procedure I, Figure 507.1-3 to determine compliance with 3.3.4.2.8.

4.6.4.2.9 Salt fog. - This test shall be performed as specified in MIL-STD-810, Method 509.1, Procedure I to determine compliance with 3.3.4.2.9.

4.6.4.2.10 Dust. - This test shall be performed as specified in MIL-STD-810, Method 510.1, Procedure I to determine compliance with 3.3.4.2.10. The temperature change during the test shall not exceed 3°F per minute.

4.6.4.2.11 Fungus.- This test shall be performed as specified in MIL-STD-810, Method 508.1, Procedure I to determine compliance with 3.3.4.2.11.

4.6.5 Performance. - The test specified in 4.6.5.1 through 4.6.5.9 inclusive shall be performed to determine compliance with 3.3.5 with the laser trainer connected as follows - A push button (depress to close type) shall be provided to simulate the machine gun trigger which will supply 24 ± 6 VDC to laser trainer input terminal when it is depressed. The negative side of the power supply shall be connected to the trainer mounting ring. A He-Ne power meter capable of measuring 2 milliwatts and having an external connection for a oscilloscope for which to measure the waveform of the laser pulse as seen by the detector, shall be provided. The oscilloscope shall be connected such that when the push button is depressed, supplying 24 ± 6 VDC to the laser trainer, the oscilloscope will trigger and the time between the applied voltage and the resulting laser pulse can be measured and the pulse duration can also be measured by the waveform shown on the screen of the oscilloscope.

NOTE: If the laser tube fails to lase or has insufficient power output allow a maximum of 30 minutes warm up time before performing 4.6.5.1 through 4.6.5.9.

Source: https://assist.dla.mil -- Downloaded: 20260-22T19:25Z
Check the source to verify that this is the current version before use.

4.6.5.1 Power input control.- To determine compliance with 3.3.5.1, the laser trainer shall be connected as specified in 4.6.5, the laser switch positioned to the "fire" position with" the mode switch to "flash" position. Depress the push button (applies 24 ± 6 VDC to laser trainer). The laser trainer shall emit only one pulse for each time the button is depressed. With the mode switch in "continuous" position, the laser trainer shall emit a continuous beam as long as the button is depressed. The laser trainer shall meet the requirement of 3.3.5.1.

4.6.5.2 Continuous or flash mode switch.- This test shall be performed In conjunction with 4.6.5.1 to determine compliance with the requirements of 3.3.5.2.

4.6.5.3 Safe and fire laser switch.- This test shall be performed in conjunction with 4.6.5.1. Position the "safe and fire" switch to the "safe" position and depress push button, the laser trainer shall not emit laser light to insure compliance with 3.3.5.3.

4.6.5.4 Trigger lamp.- This test shall be performed in conjunction with 4.6.5.1 to determine compliance with 3.3.5.3 and 3.3.5.3.1. Depressing the push button shall turn on the trigger lamp and it will remain lit until the push button is released. It shall also turn on the laser "On" lamp when the "safe and fire" switch is in the "fire" position but shall be off in the "safe" position.

4.6.5.5 Continuous mode. - This test shall be performed in conjunction with 4.6.5.1. With the trainer set up as specified in 4.6.5, the time between the depression of the push button and the emitted laser beam shall be as specified in 3.3.5.5.

4.6.5.6 Flash mode. - This test shall be performed in conjunction with 4.6.5.1. With the trainer set up as specified i 4.6.5 depression of push button shall produce one laser pulse of duration and starting times as specified in 3.3.5.6.

4.6.5.7 Repetition rate. - This test shall be performed in conjunction with 4.6.5.1. In place of the push button supply a 24V pulsed input 100 milliseconds ON and 200 milliseconds OFF for a total of ten pulses. Set the laser trainer to the "FLASH" mode and supply the ten voltage pulses to the trainer input terminal. The laser trainer shall emit ten laser pulses as viewed on the oscilloscope screen. Repeat with the trainer mode switch on "Continuous" and view the oscilloscope screen for ten laser pulses. In both conditions, all pulses shall be of at least that power (laser) as in normal operation as specified in 3.3.5.8 to determine compliance with the requirements of 3.3.5.7.

4.6.5.8 Power output. - This test shall be performed in conjunction with 4.6.5.1. A standard He-Ne power meter shall be utilized to measure the output power of the laser trainer in the continuous mode as set forth in 3.3.5.8.1 and 3.3.5.8.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

4.6.5.9 Operating input voltage.- This test shall be performed in conjunction with 4.6.4.1. The laser input voltage shall be set at 18 VDC and 30 VDC while performing tests 4.6.5.7 and 4.6.5.8. Also, the time between input voltage and the laser light emitted, as measured in accordance with 4.6.5, shall conform to the requirements of 3.3.5.9.1, 3.3.5.9.2, and 3.3.5.9.3.

4.6.6 Sealing. - This test shall be performed utilizing special test equipment. The pressure testing equipment to test sealing of the laser trainer shall conform to the internal pressure test requirements of Drawing F8565556, pressure - two station, to conform to the requirements of 3.3.6. At the conclusion of this test the trainer shall be pressurized in accordance with the requirements of 3.3.6.1.

4.6.7 Collimation. - This test shall be performed utilizing the special collimation fixture Drawing F11739406 to determine compliance with 3.3.7. With the laser trainer installed into the collimation fixture and mode switch on “Continuous” depress push button. The laser beam shall be directed on a 200 ± 20 foot target shall be parallel to the longitudinal axis of the mounting ring as specified in 3.3.7.

4.6.8 Reverse polarity protection. - This test shall be performed in conjunction with 4.6.5.1. With the input voltage reversed, negative 24 ± 6 VDC to the laser trainer input terminal and positive 24 ± 6 VDC to mounting ring (ground). The laser trainer shall not be damaged as set forth in 3.3.8.

4.6.9 Multipurpose carrying case. - Certification shall be provided to insure that the carrying case successfully met the inspection requirements of SQAP 11743778 to insure compliance with 3.3.9.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11741700 for Laser, Trainer, M55 and 10554640 for Laser, Trainer, M55 W/E. The level of protection shall be as specified in the contract.

6. NOTES

6.1 Intended use. - The Laser, Gunnery Trainer is a training device which will mount in the M73 Machine Gun Mount of the M48A3 Tank, M60 Tank, M60Al Tank and AR/AW M551 and permit year round training indoors or outdoors in the techniques of zeroing, accuracy and rapidity of laying the gun using direct fire sights. The Laser, Trainer M55 in conjunction with the Trainer, Field Artillery; Direct Fire (laser): 11746409 is used in training Howitzer Sections on the 105-MM Howitzer M102 Towed, 105mm Howitzer M101 self-propelled, 105mm Howitzer Ml14Al Towed, 155mm Howitzer M109Al self-propelled, 105mm Howitzer XM204 Towed and the 155mm Howitzer XM198 Towed.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T19:25Z
Check the source to verify that this is the current version before use.

The trainer will provide year-around training of section personnel in standard direct fire techniques while engaging both stationary and moving targets on a scale range. The trainer will allow realistic firing exercises to be conducted in garrison, without having to move howitzers to a live firing range and without expending either sub caliber or service ammunition.

6.2 Ordering Data.- Procurement documents should specify the following:

a. Title, number and date of specification
b. Applicable packaging data sheet (see 5.1)
c. Selection of applicable levels of preservation, packaging and packing.
d. Applicable stock number
e. Requirements for First Article submission which may include additional tests selected by the procuring agency.

Custodian

Army - MU

Preparing activity:

Army - MU

Project Number 1240-A682

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Frankford Arsenal
ATTN: SARFA-MDM
Philadelphia, PA 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-L-48407B(MU) Laser, Gunnery Trainer: M55

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:25Z
Check the source to verify that this is the current version before use.

NOTICE OF INACTIVATION FOR NEW DESIGN

INCH-POUND

MIL-L-48407B(AR)
NOTICE 2
29 November 1995

MILITARY SPECIFICATION

LASER, TANK GUNNERY TRAINER: M55

This notice should be filed in front of MIL-L-48407B(AR), dated 31 May 1976, with Validation Notice 1, dated 16 May 1988.

MIL-L-48407B(AR) dated 31 May 1976, with Validation Notice 1, dated 16 May 1988 is inactive for new design and is no longer used, except for replacement purposes.

Preparing Activity:
Army - AR

AMSC N/A

FSC 1240

DISTRIBUTION STATEMENT A. Approved for public rclcase; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-22T19:25Z
Check the source to verify that this is the current version before use.